

Yihan Li

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Education

Sun Yat-sen University	Sep 2023 – Jul 2027 (Expected)
<i>B.A. in English Language & Literature, School of Foreign Languages</i>	<i>Major GPA: 3.8/4.0</i>
<i>Minor in Intelligent Unmanned Systems</i>	<i>GPA: 4.0/4.0</i>
<i>Minor in Computer Science & Technology</i>	<i>GPA: 3.8/4.0</i>

Publications

Anticipating Contact for Visuo-Tactile Diffusion Policies	under review
Yating Feng, Yuhang Zheng, Lixin Xu, Yihan Li , Jiaju Yin, Renjing Xu	
Goldsmith: Gold-Loss-Guided Prompt Training for Agentic Annotation	under review
Yihan Li , Hanyi Zhang, Xiaoxi Jiang, Man Guo	
Unordered Landmark Visual Navigation	under review
Hao Ren, Junzhe Zhu, Yihan Li , Zetong Bi, Le Zheng, Zhi Li, Yiqing Yuan, Zhaoliang Wan, Lu Qi, Hui Cheng	
OmniTrav: A Dataset and Benchmark for 360° Vision-based Traversability via Foundation Models	under review
Yihan Li , Hao Ren, Zhenglan Jiang, Junzhe Zhu, Hui Cheng	
RAPID Hand: A Robust, Affordable, Perception-Integrated, Dexterous Manipulation Platform for Generalist Robot Autonomy	NeurIPS 2025
Zhaoliang Wan, Zetong Bi, Zida Zhou, Hao Ren, Yiming Zeng, Yihan Li , Lu Qi, Xu Yang, Ming-Hsuan Yang, Hui Cheng	

Work Experience

HCL Lab (HKUST(GZ))	Jan 2026 – Present
<i>Research Assistant</i>	<i>Prof. Renjing Xu</i>
- Conducting dexterous-hand reinforcement learning research, with emphasis on Real-to-Sim transfer pipelines for reproducible policy training and evaluation. - Developing and validating VLA-style multimodal manipulation methods for visuo-tactile dexterous tasks.	
Yun Drone	Jun 2025 – Mar 2026
<i>UAV Development Intern</i>	
- Selected UAV propulsion components and participated in formation-flight development, navigation and localization, and flight-test workflow design. - Explored RL-based disturbance-resilient control and analyzed flight logs to support anti-disturbance tuning and system iteration.	
RAPID Lab (SYSU)	Nov 2024 – Jun 2025
<i>Research Assistant</i>	<i>Prof. Hui Cheng</i>
- Developed the dexterous-hand control stack, visuo-tactile temporal alignment, and MuJoCo Sim2Real validation workflow for manipulation experiments. - Built visual obstacle-avoidance and topological-map planning models, connecting perception outputs with embodied navigation tasks.	
RoboTech Association (SYSU)	Sep 2024 – Present
<i>President; Competition Team Captain</i>	
- Led 160+ members and competition teams, coordinating training schedules, technical reviews, and cross-group collaboration. - Authored robotics onboarding tutorials across mechanical design, hardware, embedded control, and algorithms.	

Projects

RoboMaster Visual System Development (RMUL)

Mar 2025 – Jul 2025

Vice Captain; Team Advisor; Algorithm Group Leader for Navigation and Vision

- Built Jetson/NUC vision with EKF auto-aiming (90%+ hit rate) and TensorRT/OpenVINO YOLO inference at 10 ms.
- Led navigation and vision algorithm planning, STM32 communication, and closed-loop deployment for competitive robotics.

RAPID Hand: Perception-Integrated Dexterous Manipulation Platform

Nov 2024 – Jun 2025

Research Assistant

- Implemented drivers, visuo-tactile fusion, MuJoCo Sim2Real training, and Diffusion Policy tasks with 50/50 success.
- Integrated 1 kHz hand control, wrist vision, fingertip tactile sensing, and joint proprioception under 7 ms synchronization latency.

OmniTrav: 360° Vision-based Traversability Benchmark

Jun 2025 – Mar 2026

First Author

- Proposed C2T RGB traversability and built a 4.7k-scene, 250k-frame, 577 GB automated foundation-model dataset.
- Evaluated pinhole, fisheye, and panoramic cameras; achieved panoramic obstacle detection F1 of 1.0 and pinhole median error of 0.27 m.

GenreShift and Rosetta: LLM Systems for Linguistic Research

Oct 2024 – Nov 2025

Project Leader

- Built GenreShift (corpus, LoRA, MoE, patent No. 2025105553019) and Rosetta (Streamlit/Docker annotation with Kimi/DeepSeek APIs).
- Supported interactive concept management, data import/export, and LLM-assisted academic discourse annotation.

Awards

- First Prize, National Level, China Robot and Artificial Intelligence Competition, 2025-08.
- Second Prize, National Level, National Undergraduate Electronics Design Contest, 2025-08.
- Second Prize, National Level, National Undergraduate Robotics Competition (RoboMaster / RoboCon), 2025.
- First Prize, National Level, National Intelligent Unmanned Systems Application Challenge, 2025-07.

Skills

Programming:	ROS1/2, Python, C/C++, MATLAB, STM32/Arduino.
Robotics/ML:	Visual navigation, 3D perception, dexterous manipulation, MuJoCo, TensorRT, OpenVINO, LLM annotation systems.
Engineering and Languages:	Fusion 360, SolidWorks, EDA design; AOPA Multi-rotor VLOS Pilot; Mandarin (native), English (TEM-4), French.